INCH-POUND
MIL-DTL-49285/3B
2 March 2018
SUPERSEDING
MIL-DTL-49285/3A
15 May 2009

DETAIL SPECIFICATION SHEET
CABLE, SPECIAL PURPOSE, ELECTRICAL, NINE INDIVIDUALLY SHIELDED PAIRS OF 22 AWG SOLID WIRE

Inactive for new design after 16 June 1997.

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-49285.

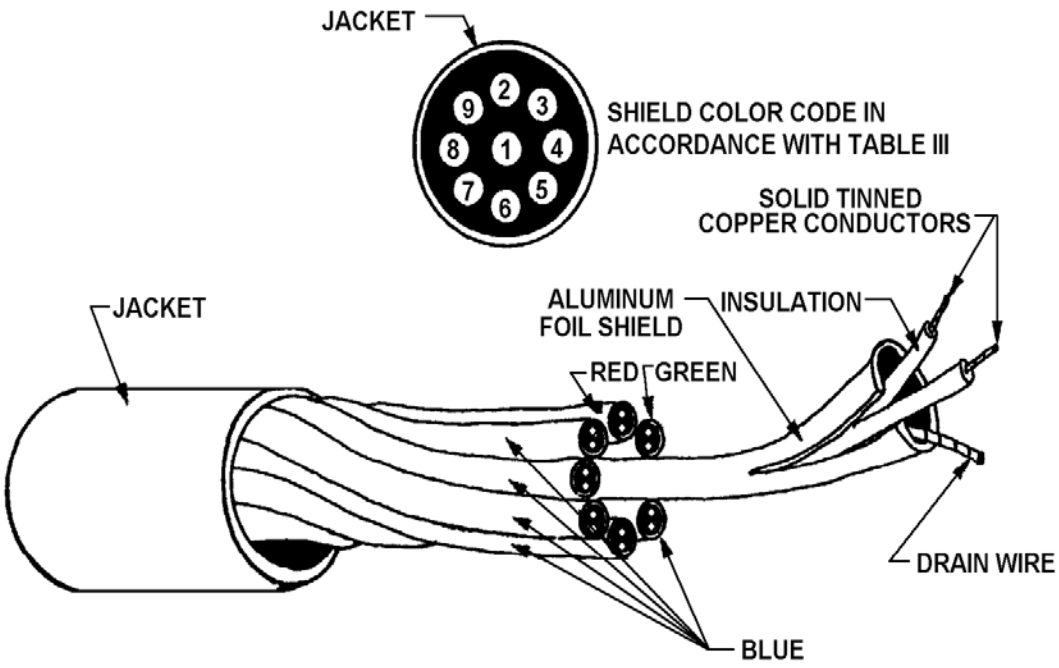


FIGURE 1. Cable illustration.

Engineering information: See table I.

Note: Not recommended for shipboard or aerospace applications due to toxic smoke release in the event of a fire. PVC is not recommended in other applications (i.e. confined spaces, low ventilation) where there is the potential for human smoke exposure in a fire.

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TABLE I. Description, electrical.

Electrical characteristics	Requirements
Nominal capacitance between conductors	40 pF/ft (picofarads/foot)
Nominal capacitance between the conductor and the shield with one conductor tied to the shield	77 pF/ft
Continuous working voltage	300 V rms maximum between conductors
Nominal capacitance between adjacent shields	150 pF/ft
Continuous working voltage between adjacent shields	50 volts maximum
Shield dc resistance	14 ohms/1,000 feet maximum

TABLE II. Description, physical.

Components	Construction details
Number of pairs	Nine individually shielded
Conductor type and wire size	Stranded tinned copper 22 AWG
Conductor insulation	Polyvinyl chloride
Conductor insulation thickness	0.013 inch nominal thickness
Drain wire type and size	Stranded tinned copper 22 AWG
Jacket material	Polyvinyl chloride
Jacket thickness	0.038 inch
Finished cable diameter	0.440 inch maximum outside diameter
Cable style (UL)	2464
Tensile strength (jacket)	2,000 pounds per square inch minimum
Elongation (jacket)	150 percent minimum
Overall cabling lay length	1.3 twists per foot ± 10 percent

Cabled pair orientation: The position of the red and green pair shield shall determine the rotation direction for pair location (see figure 1).

Shield color code. See table III.

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TABLE III. Shield color code.

Pair numbers	Shield color
1st pair	Red center core
2nd pair	Red
3rd pair	Green
4th pair	Blue – 1
5th pair	Blue – 2
6th pair	Blue – 3
7th pair	Blue – 4
8th pair	Blue – 5
9th pair	Blue – 6

REQUIREMENTS:

Design and construction. See table II.

Shield location and orientation. The polyester aluminum shield is to be located on the outer circumference of the pair, with the aluminum foil side inward. Insulation on the exterior of the shield is to be complete with no aluminum available for contact.

Drain wire location. The drain wire is to be spirally located beneath the shield with the insulated conductors. It is to be in continuous contact with the aluminum surface of the foil shield throughout the cable.

Cable temperature rating. The cable temperature rating shall be -20°C to +80°C.

Flammability. The cable shall pass UL 2556 flame test requirements.

Shield integrity test.*

Frequency	Response
100 kHz	114 dB or more below reference
500 kHz	105 dB or more below reference
1 MHz	99 dB or more below reference
5 MHz	85 dB or more below reference
10 MHz	79 dB or more below reference
15 MHz	74 dB or more below reference
20 MHz	70 dB or more below reference
25 MHz	65 dB or more below reference
30 MHz	61 dB or more below reference
100 MHz	42 dB or more below reference

* Any two readings may deviate to a value of 0.9 times the stated limits.

Fixture resonance will occur between 40 and 70 MHz.

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Crosstalk test limits. All values are minimums.

Frequency (kHz)	Near end crosstalk (dB)	Far end crosstalk (dB)
40	58	58
70	56	55
100	54	53
1,000	55	52

Insulation resistance between shields. 100 megohms-1,000 feet minimum.

Durometer hardness. The cable jacket shall have a “Shore A” hardness of 82 ± 5.

Part or Identifying Number (PIN). The PIN shall be M49285/03.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

Referenced documents. In addition to MIL-DTL-49285, this document references the following: UL 2556.

CONCLUDING MATERIAL

Custodians:
Army – CR
Navy – SH
Air Force – 85
DLA - CC

Preparing activity:
DLA-CC

Review activities:
Army – MI
Air Force – 99

(Project 6145-2018-011)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change. You should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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